

Lab 4

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In [1]:

```
import matplotlib.pyplot as plt
import cv2

img = cv2.imread('images/Parrot.png', cv2.IMREAD_GRAYSCALE)
img = cv2.resize(img, (200, 200))

threshold_values = [0, 50, 100, 150, 200]

for threshold_value in threshold_values:
    ret, thresh = cv2.threshold(img, threshold_value, 255, cv2.THRESH_BINARY)
    plt.imshow(thresh, cmap='gray', vmin=0, vmax=255)
    plt.title(f'Thresholded Image ({threshold_value})')
    plt.axis('off')
    plt.show()

plt.imshow(img, cmap='gray')
plt.title('Original Image')
plt.axis('off')
plt.show()
```

Thresholded Image (0)

Thresholded Image (50)



Thresholded Image (100)



Thresholded Image (150)



Thresholded Image (200)



Original Image



In [2]:

```
import matplotlib
import matplotlib.pyplot as plt
import numpy as np

from skimage import data, img_as_float
from skimage import exposure
```

In [3]:

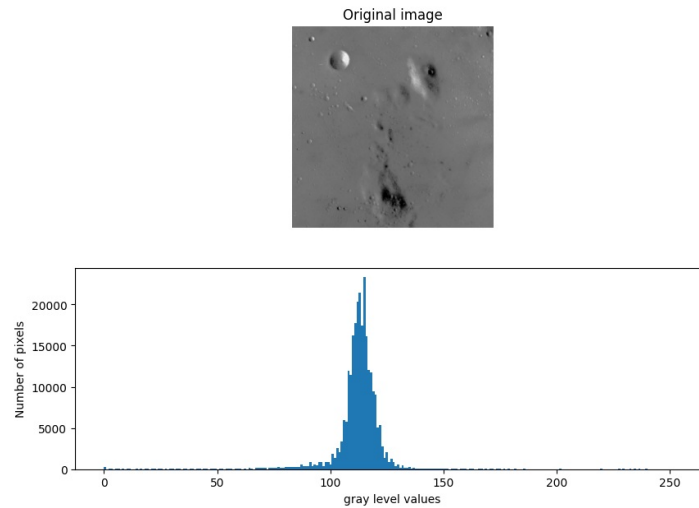
```
img = data.moon()
```

In [4]:

```
p2, p98 = np.percentile(img, (2, 98))
img_rescale = exposure.rescale_intensity(img, in_range=(p2, p98))
```

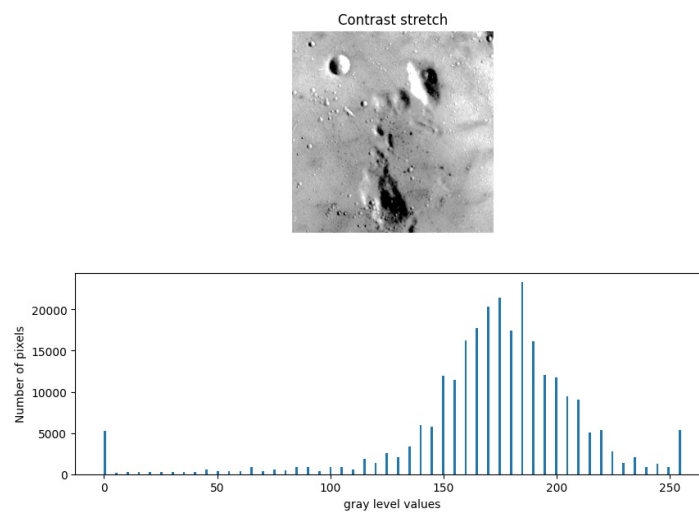
In [5]:

```
fig = plt.figure(figsize=(10, 7))
fig.add_subplot(2, 1, 1)
plt.imshow(img, cmap='gray')
plt.axis('off')
plt.title('Original image')
fig.add_subplot(2, 1, 2)
plt.hist(img.flat, bins=256, range=(0, 255))
plt.xlabel('gray level values')
plt.ylabel('Number of pixels')
plt.show()
```



In [6]:

```
fig = plt.figure(figsize=(10, 7))
fig.add_subplot(2, 1, 1)
plt.imshow(img_rescale, cmap='gray')
plt.axis('off')
plt.title('Contrast stretch')
fig.add_subplot(2, 1, 2)
plt.hist(img_rescale.flat, bins=256, range=(0, 255))
plt.xlabel('gray level values')
plt.ylabel('Number of pixels')
plt.show()
```



In [7]:

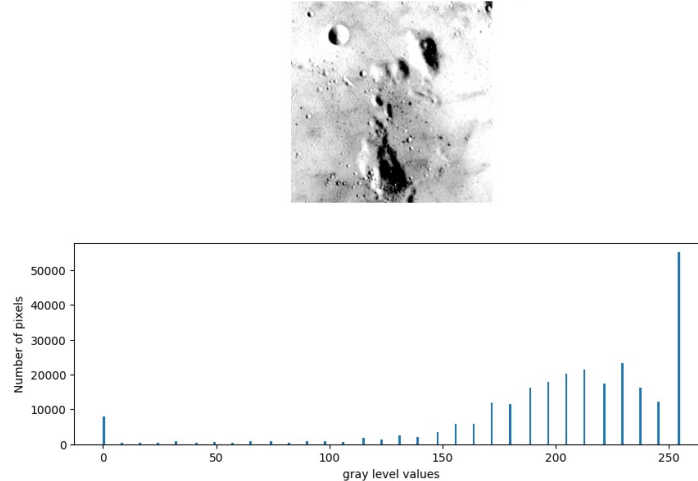
```

p3, p80 = np.percentile(img, (3, 80))
img_rescale2 = exposure.rescale_intensity(img, in_range=(p3, p80))

fig = plt.figure(figsize=(10, 7))
fig.add_subplot(2, 1, 1)
plt.imshow(img_rescale2, cmap='gray')
plt.axis('off')
plt.title('Contrast stretch (3rd - 80th percentile)')
fig.add_subplot(2, 1, 2)
plt.hist(img_rescale2.flat, bins=256, range=(0, 255))
plt.xlabel('gray level values')
plt.ylabel('Number of pixels')
plt.show()

```

Contrast stretch (3rd - 80th percentile)



In [8]:

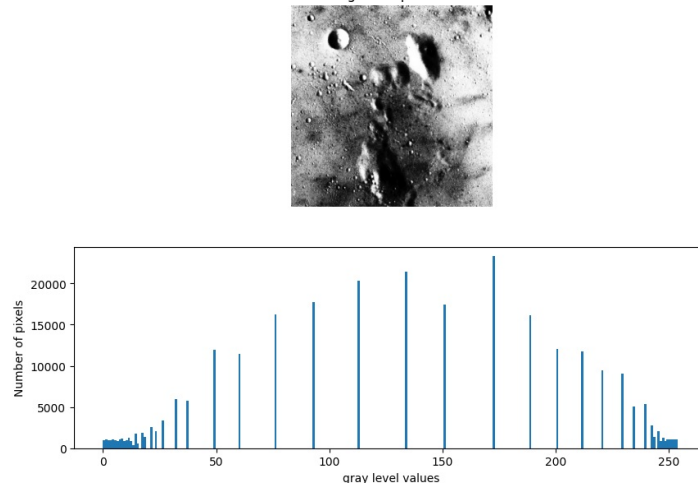
```

img_eq = exposure.equalize_hist(img)

fig = plt.figure(figsize=(10, 7))
fig.add_subplot(2, 1, 1)
plt.imshow(img_eq, cmap='gray')
plt.axis('off')
plt.title('Histogram equalization')
fig.add_subplot(2, 1, 2)
plt.hist((img_eq * 255).astype('uint8').flat, bins=256, range=(0, 255))
plt.xlabel('gray level values')
plt.ylabel('Number of pixels')
plt.show()

```

Histogram equalization



In [9]:

```
from skimage.exposure import match_histograms

reference = data.rocket()
image = data.chelsea()
matched = match_histograms(image, reference, channel_axis=-1)

fig, axes = plt.subplots(1, 3, figsize=(12, 4))
for ax, im, t in zip(axes, (image, reference, matched), ('Chelsea', 'Rocket (reference)', 'Matched')):
    ax.imshow(im)
    ax.set_title(t)
    ax.axis('off')
plt.tight_layout()
plt.show()
```

